

Manual

Detailed Planning with Production Variants HLS-FFV 8.1

Version 1.0.23049

Last changed on: 01.09.2020

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1 Overview – Detailed Scheduling with Production Variants

Purpose

This function package allows the management of relationships between article, tool and machine and their consideration during assignment in the HYDRA shop floor scheduling (HLS).

You use the function package when

- You wish to ensure that due to the article to be produced, certain orders are manufactured only at specific workplaces or with specific tools,
- Default values (e.g. setup time, target cycle) change, depending on the use of a tool or scheduling at a specific workplace

Integration

Production variants are taken into consideration in the graphic planning board both for manual scheduling and automatic assignment.

Features

- Editing function for management of relationships between article, tool and machine.
- Selective control of the processing for specific order types or for specific machine/ workplace groups.
- Consideration of the production variants (tool selection or plausibility check) during the detailed scheduling of operations in the HYDRA shop floor scheduling.
- Possibility of uploading production variants via an interface (customizing) and of manual processing in MOC.
- Optional: Consideration of the production variants (tool selection or plausibility check) even during the transfer of order sequencing data via the ERP/PPS interface

2 Processing Instructions in Conjunction with Production Variants

Information on activation of the functions can be found in the following documents:

- Order types, document [MOC_MDOrdertypes.pdf](#)
- Groups, document [MOC_Resourcegroups.pdf](#)

Processing instructions in conjunction with production variants can be found in the following documents:

- Graphic planning board, document [MOC_GraphicPlanning.pdf](#)
- Automatic scheduling, document [MBL_AutomaticPlanning.pdf](#)

3 Production Variants

Overview

Menu	Master data → Production control → Production variants
Transaction code	prodvar
Function authorization	prodvar

Purpose

You can use the function to create or modify production variants in the system.

Integration

You can use production variants to define different options for specific articles. The system integrates these options during planning in HYDRA Shop Floor Scheduling (HLS). You can define different production variants for one article/item. An alternative production variant must exist for the article, if the order type specifies that production variants are used for planning. Otherwise, you cannot plan the operation for the machine.

If you do not want to use production variants for all areas, they can be "skipped" for single groups. Consequently, the [Group](#) configuration takes priority over the order type configuration.

An exceptional case occurs if "N" - no utilization" is configured for the order type. In this case, production variants are generally not used for this order type. The group configuration does not have any effect in this case.

Selection criteria

The application provides the following selection criteria:

Article

The system searches for production variants available for the entered article/item.

Workplace

Number of the workplace / machine where you can manufacture the article with the resource.

Workplace group

Number of the capacity group where you can manufacture the article.

Resource

Resource (number) you can use to manufacture an article as part of a production variant.

If you want to include a resource (e.g. tool) in checking for valid production variants and in planning, you should specify the resource here, otherwise leave the field empty. You cannot define a production variant at the same time for both a resource and a resource family.

Resource family

The system does not support this field when identifying production variants. You can merely use it for comments.

Status (blocked/released)

Possible values describing the current status of a production variant.

Only one version at a time may have a status value of *Released* per production variant. You can use this field to filter by released or blocked production variants.

Detail application "production variants"

The table view of the detail application offers an overview of existing entries.

Field description**Article**

Article number of the article to be manufactured.

The article number length is restricted to 15 characters if HYDRA ALS (interfacing to the ARBURG host computer system) is used.

Article designation/name

Name of the article.

Group

Number of the workplace group where the article can be manufactured with the tool.

Complete this field in any case, even if the production variant should only be based on the combination of *Article* and **Workplace**.

Workplace

Number of the workplace / machine where you can manufacture the article with the resource.

Leave this field empty, if the production variant should only be based on the combination of *Article* and **Group**.

Resource type

Type of the resource you can use to manufacture an article as part of a production variant. The system only supports the resource type "WNR".

Resource

Resource (number) you can use to manufacture an article as part of a production variant.

If you want to include a resource (e.g. tool) in checking for valid production variants and in planning, you should specify the resource here, otherwise leave the field empty. You cannot define a production variant at the same time for both a resource and a resource family.

This is a key field if the interfacing to the ARBURG host computer system (ALS) is used. The resource number length (tool number) is restricted to 15 characters in this case.

Resource family

The system does not support this field when identifying production variants. You can merely use it for comments.

Please observe the following restrictions:

- If you create a new production variant and assign a resource, the resource family is not automatically taken from the pool of resources.
- The system does not check whether the resource family is valid. In case of doubt, you can select the resource family via the search dialog.
- If you change a resource family for a resource in the pool of resources, the system does not synchronize this with the production variant.

Number of resources

The number of resources required for the production variant.

Machine/operator relation for setup/production including qualifications

Define the personnel requirements that are needed for setting up or producing the operation including the relevant qualification.

You cannot use a production variant to reset (set to zero) the workforce requirements previously defined for the OP.



You can define the machine/operator relation for production variants as of SP13 and HLS-FFV 8.2. Existing customers who want to use the fields must execute the database patch provided by SP13.

Priority

Priority for identifying the production variant. A higher value stands for a higher priority.

Version

You can use this field (as part of the key) to assign version numbers to production variants. Enter "1" at the very left of the field if versioning is not used.

Only one version of a production variant at a time may have the "released" status (cf. *Status* field description).

Target cycle

Target cycle for machine monitoring in MDE. This value is defined per 1000 machine cycles.

The system keeps the target cycle defined for the operation, if you enter 0 in the "target cycle" field.

Admissible deviation

Reserved.

Partitioning

Number of the quantity produced during one machine cycle. For each machine cycle, the system enters/posts a quantity produced that corresponds to the partitioning value.

The system keeps the partitioning defined for the operation, if you enter 0 in the "partitioning" field.

Setup time

When planning in HYDRA Shop Floor Scheduling, the system uses this value instead of the default setup time value defined for the operation.

Please note: If setup time is stored as a formula for the operation, the setup time value is recalculated each time you change the group manually or you update the order. In this way, the setup time value entered here can be overwritten again.

Teardown/retooling time

When planning in HYDRA Shop Floor Scheduling, the system uses this value instead of the default teardown/retooling time value defined for the operation.

Please note: If retooling/teardown time is stored as a formula for the operation, the retooling/teardown time value is recalculated each time you change the group manually or you update the order. In this way, the retooling/teardown time value entered here can be overwritten again.

Valid from

Date from which the production variant is valid.

Valid until

Date up to which the production variant is valid. Enter 31.12.9999 here if the production variant should be valid until further notice. The system does not update the "valid until" date if the status is changed.

Status

Possible values describing the current status of a production variant:

F	Released
S	Blocked

Only one version at a time may have a status value of *Released* per production variant. If you create a new production variant with the *released* status or if you change the status of an existing variant to *released*, HYDRA checks whether an existing production variant has already the *released* status. If so, the existing version is automatically set to the *blocked* status.

Blocking reason

You can enter a numeric blocking reason here if a production variant is set to the "blocked" status.

Comment

Comment field for this production variant.

Toolbar



Generate order

Use the "generate order" function to create orders from work plans based on the specified configuration.

Activation of production variants

You can configure in the order type if you want to use production variants. Select the order type and modify the entry for "consideration of production variants in planning".

Consideration of production variants in planning:

- Identification (E): you can select from existing production variants.
- Checking only (P): the system only checks whether a valid production variant exists.
- No use (N): the system does not check the production variants.